

Skills Summary

Naming a Conic and Reading Its Key Features

Use this reference while completing your worksheet or when studying.

Skill 1 — Name the family before anything else

General form: $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$. Move every term to one side first, then read A , B , C straight off.

Indicator: $B^2 - 4AC$.

- negative and $A = C$: **circle** • negative and $A \neq C$: **ellipse**
- zero (only one squared term): **parabola** • positive (squares of opposite sign): **hyperbola**

Example: Name the conic $16x^2 - y^2 = 16$.

Step 1. Only the family is wanted, and the family is decided by $B^2 - 4AC$ alone — so read off A , B , C and compute nothing else.

Step 2. One side: $16x^2 - y^2 - 16 = 0$, so $A = 16$, $B = 0$, $C = -1$, and $B^2 - 4AC = 0 - 4(16)(-1)$.

$\Rightarrow$ **64**, positive, so it is a **hyperbola**

Try it: Name the conic $x^2 + 9y^2 = 9$.

check: $-$, negative with $A \neq C$, an ellipse

Skill 2 — Circle: centre and radius

From $x^2 + y^2 + Dx + Ey + F = 0$ **to** $(x - h)^2 + (y - k)^2 = r^2$: complete the square on the x terms and on the y terms.

Shortcut: $h = -D/2$, $k = -E/2$, $r = \sqrt{h^2 + k^2 - F}$.

The bracket signs are *minus* h and *minus* k , so $(y+3)^2$ means $k = -3$. And the right-hand side is r^2 — take the square root before writing the radius.

Example: Centre and radius of $x^2 + y^2 - 4x + 6y - 3 = 0$.

Step 1. Both squares are present with equal coefficients, so it is a circle and the shortcut formulas apply directly — no need to complete the square by hand.

Step 2. $h = -(-4)/2$, $k = -6/2$, $r = \sqrt{h^2 + k^2 - F} = \sqrt{4 + 9 + 3} = \sqrt{16}$.

$\Rightarrow$ centre **(2, -3)**, radius **4**

Try it: Centre and radius of $x^2 + y^2 + 2x - 8y - 19 = 0$.

check: centre **(-1, 4)**, radius **6**

Skill 3 — Parabola: axis and vertex

From $y = ax^2 + bx + c$: the axis of symmetry is the vertical line $x = -b/(2a)$; the vertex is that x together with the y you get by substituting it back. Opens **up** when $a > 0$, **down** when $a < 0$.

The x -intercepts (if any) come from solving $ax^2 + bx + c = 0$, and the axis always sits exactly halfway between them — a free check on both answers.

Example: Axis and vertex of $y = x^2 - 8x + 12$.

Step 1. The vertex is the one point on the axis of symmetry, so find the axis first and the vertex falls out of it — no completing the square needed.

Step 2. $a = 1$, $b = -8$, so $x = -(-8)/2 = 4$; then $y = 4^2 - 8(4) + 12 = 16 - 32 + 12$.

$\Rightarrow$ axis $x = 4$, vertex $(4, -4)$, opening up

Try it: Axis and vertex of $y = x^2 + 6x + 11$.

check: axis $x = -3$, vertex $(-3, -2)$

Skill 4 — Ellipse: vertices and foci

Standard form: $x^2/a^2 + y^2/b^2 = 1$ with the centre at the origin. The denominators are squares of the distances, so take square roots before plotting.

Vertices $(\pm a, 0)$, co-vertices $(0, \pm b)$, foci $(\pm c, 0)$ with $c^2 = a^2 - b^2$ — when the larger denominator sits under x^2 , all of this runs along the x -axis. A hyperbola $x^2/a^2 - y^2/b^2 = 1$ uses the same a for its vertices, but $c^2 = a^2 + b^2$ instead.

Example: Vertices and foci of $x^2/25 + y^2/16 = 1$.

Step 1. Every key feature here is a distance, and the equation stores distances as squares — so the whole job is three square roots.

Step 2. $a = \sqrt{25}$, $b = \sqrt{16}$, $c = \sqrt{25 - 16} = \sqrt{9}$.

$\Rightarrow$ $a = 5$, $b = 4$, $c = 3$: vertices $(\pm 5, 0)$, foci $(\pm 3, 0)$

Try it: Find a , b and c for $x^2/100 + y^2/64 = 1$.

check: $a = 10$, $b = 8$, $c = 6$